Projecting Hitters at Altitude

A Coors-aware Marcel projection system, backtested on 2021–2025 and checked against the 2026 season

Dan Tunningley · October 2026

Park adjustment fixes newcomers; returning Rockies are the open problem

15 → 0

points of wOBA: bias on hitters joining the Rockies, plain Marcel vs park-adjusted (2021–25 backtest)

2.46M

plate appearances behind the model: every regular-season MLB game, 2012–2025

–17

points: how far the park-adjusted model overrated returning Rockies hitters in 2026

1 point = .001 wOBA. The same pattern showed up in the 2021–2025 backtest and again in the 2026 season.

Coors-aware Marcel projections · Data: Retrosheet 2012–2025, Fox Sports 2026

Three versions, each adding one layer to the last

1

Plain Marcel

Tom Tango's baseline: last three seasons weighted 5/4/3, regressed toward league average, adjusted for age. No park handling.

→

2

Park-adjusted

Neutralizes each past plate appearance by its park's factor for each outcome, projects in a neutral park, then places it back in the parks ahead.

→

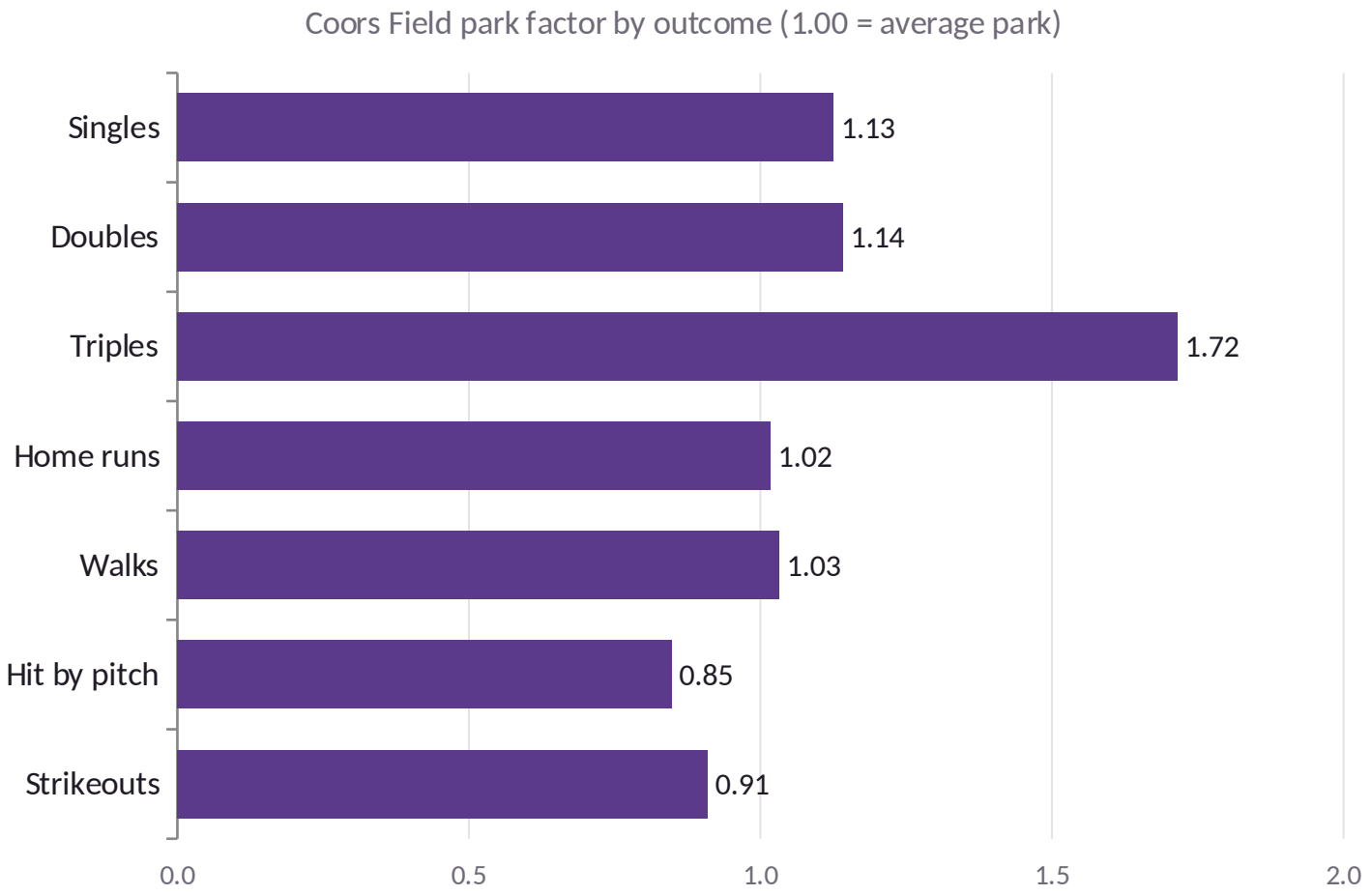
3

Park + Rockies multipliers

Adds Rockies-specific home and road corrections learned from training-season errors: the altitude edge and the "Coors hangover".

Tested fairly: tuned on 2015–2019, scored on 2021–2025 seasons it never saw, then checked against actual 2026 results. Every number for a season uses only earlier seasons.

Coors inflates triples, singles and doubles far more than home runs



Triples up 72%, singles 13%, doubles 14%

The big outfield and thin air reward balls in play.

Home runs up only 2%

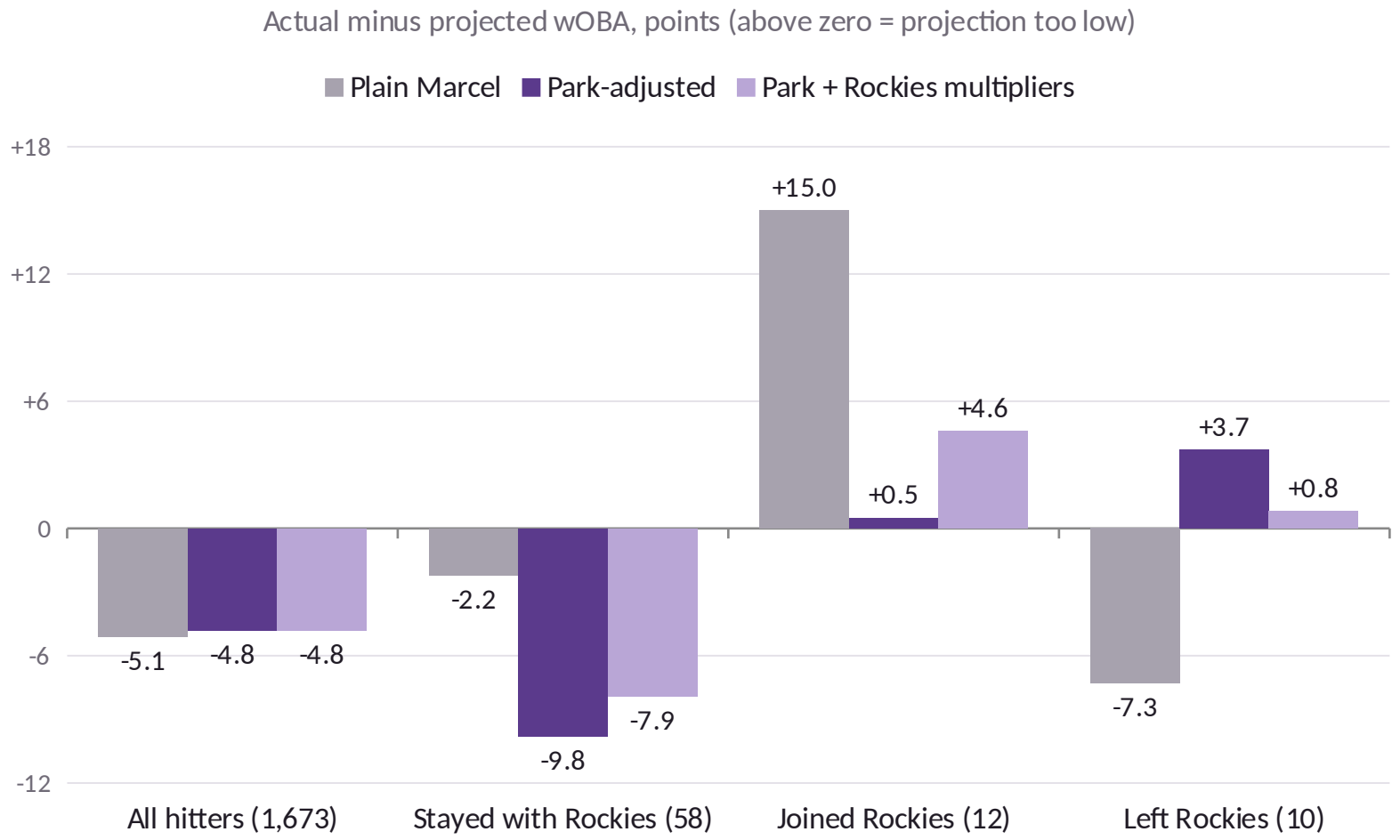
Since the humidor, Coors is not mainly a home-run park.

Why it matters

One overall park factor hides this, so the model adjusts each outcome separately.

Coors-aware Marcel projections · Data: Retrosheet 2012–2025, Fox Sports 2026

In the backtest, park adjustment removed the bias on hitters who changed teams



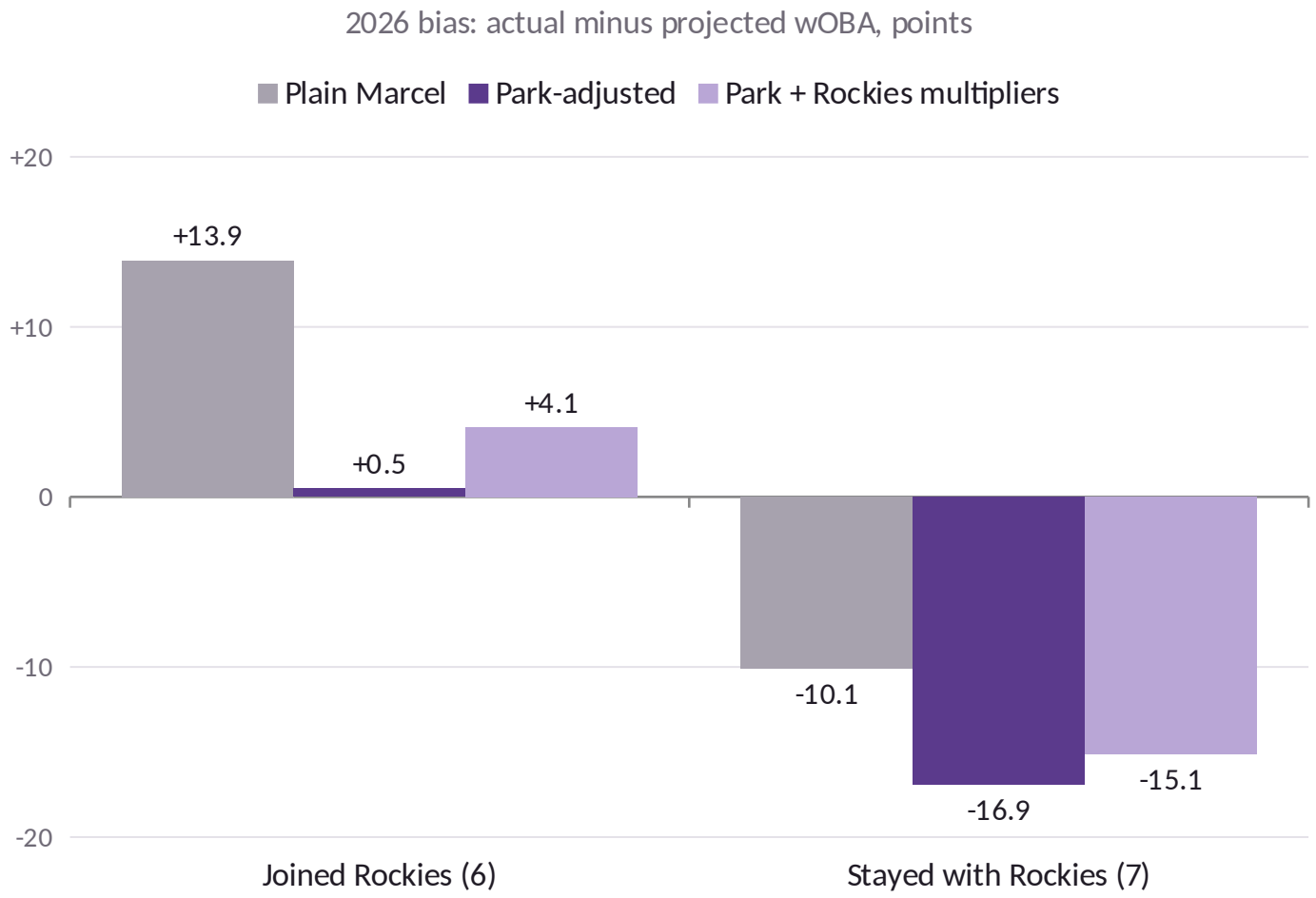
Newcomers: +15 → +0.5
Plain Marcel underrated hitters moving to Denver and overrated those leaving.

Holdovers got worse
The park-adjusted model overrated Rockies who stayed by about 10 points.

Most error is luck
All versions miss by ~.032 wOBA; luck alone would cause ~.024.

Coors-aware Marcel projections · Data: Retrosheet 2012–2025, Fox Sports 2026

2026 repeated the backtest: newcomers right, holdovers overrated



Newcomers: +0.5 points
The park-adjusted miss matched the backtest exactly. Plain Marcel would have been 14 points low.

Holdovers: −17 points
Every version overrated returning Rockies, driven by young players who regressed.

Small sample
13 hitters; rookies Rumfield and Carrigg had no MLB history to project.

Coors-aware Marcel projections · Data: Retrosheet 2012–2025, Fox Sports 2026

The biggest misses were young holdovers who got worse

Joined the Rockies

Hitter	Projected	Actual	Miss
Connor Norby	.322	.367	**+45**
Jake McCarthy	.315	.344	**+29**
Willi Castro	.322	.326	**+4**
Troy Johnston	.328	.325	**-3**
Brett Sullivan	.308	.283	**-25**
Edouard Julien	.321	.283	**-38**

Stayed with the Rockies

Hitter	Projected	Actual	Miss
Hunter Goodman	.336	.354	**+18**
Kyle Karros	.318	.334	**+16**
Mickey Moniak	.328	.318	**-10**
Tyler Freeman	.312	.283	**-29**
Brenton Doyle	.301	.258	**-43**
Ezequiel Tovar	.320	.268	**-52**
Jordan Beck	.315	.246	**-69**

2026 wOBA projected by the park + Rockies model vs actual; miss in points (1 point = .001). Misses of 30+ points below projection in orange.

Next: fix the holdover bias, then broaden the model

1

Regress toward a role-based mean

Pull hitters toward peers with similar playing time, not league average; targets the holdover overprojection directly.

2

Re-baseline to the run environment

Anchor to the latest season's league rates; removes the ~5-point league-wide overprojection.

3

Add Statcast inputs

Exit velocity, launch angle, sprint speed and spray direction likely interact with Coors in ways a flat multiplier misses.

4

Extend to pitchers

The same outcome-by-outcome framework applies, and Coors matters even more for pitcher evaluation.